Lower Quality of Life in Vitamin D-Deficient Ehlers-Danlos Syndrome Patients: Evidence from a Large National Cohort

School of Life Sciences
Arizona State University

Adair Conlin[1] | Lydia Luck[1] | Jade Reilly[1] | Jenn D. Hardy[1] | Scarlett Bernhardt[2] | Shayna Tatum[1] | Xena Eriksen[3] | Daisy Strobel[1] | Susan Holechek, Ph.D.[1,4]

[1]School of Life Sciences, Arizona State University, Tempe, Arizona, USA
[2]School of Interdisciplinary Forensics, Arizona State University, Tempe, Arizona, USA
[3]Mathematics, Computing, and Physics, Salem College, Winston Salem, North Carolina, USA
[4]Simon A. Levin Mathematical, Computational, and Modeling Sciences Center, Arizona State University, Tempe, Arizona, USA

Introduction

What is Ehlers-Danlos Syndrome (EDS)?

- An inherited genetic disorder affecting various body systems, often caused by defects in collagen formation (Neto et al., 2024).
- Abnormal collagen synthesis results in reduced bone strength and tissue elasticity.
- Symptoms include chronic pain, joint instability, fatigue, and impaired wound healing (Klett & Diegelmann, 1997).

Why vitamin D deficiency (VDD)?

- Vitamin D is vital for bone health and immune system regulation (Klett & Diegelmann, 1997).
- Symptoms of VDD overlap with those of EDS, and VDD is a frequently co-morbid in EDS patients. (Steiner et al., 2023, Ullian et al., 2023).
- Characteristics of EDS, such as gastrointestinal issues, difficulty swallowing, and disordered eating, can limit vitamin D intake.
- Supplementing vitamin D is a safe, low-cost treatment for VDD.

Research Question: Is there a meaningful association between quality of life and vitamin D deficiency in Ehlers-Danlos Syndrome patients?

Hypothesis: Vitamin D deficiency correlates with a lower quality of life in Ehlers-Danlos Syndrome patients.

Materials and Methods

Within the *All of Us* database, we identified a total cohort of 1,233 EDS patients. This was subsequently divided into two distinct groups, a cohort of 446 individuals with VDD and a comparative cohort of 787 individuals with EDS and normal vitamin D levels.

Quality of Life (QoL) Rankings

- The dataset utilized *All of Us* Quality of Life ratings Poor, Fair, Good, Very Good, and Excellent, which we categorized for each group.
- To comply with *All of Us* guidelines for minimum cohort sizes and ensure meaningful statistical comparisons, those who responded with "very good" or "excellent" were combined into a singular category, resulting in a four-level ordinal scale.

Statistical Tests

Statistical modeling and data visualization were done using R, with statistical significance defined by an alpha level of 0.05.

- Mann-Whitney U Test
  - A non-parametric test that compares the overall shift in median QoL ratings between the deficient and non-deficient study groups.
- Cochran-Armitage Test
  - Identifies a linear trend in proportions of Vitamin D deficiencies and QoL ratings.
- Ordinal Logistical Regression
  - A Proportional Odds Model that calculates an Odds Ratio and 95% Confidence Intervals was fit to the data to quantify the likelihood of a patient reporting a higher QoL based on their Vitamin D status.

Results

Our findings indicate a measurable association between Vitamin D deficiency and lower self-reported quality of life among participants with the connective tissue disorder Ehlers-Danlos syndrome.

Table 1. **Vitamin D Status by Quality of Life** - This table shows the two *All of Us* cohorts included in our research as well as the frequencies of the responses on the quality of life question.

Quality of Life	Vitamin D Deficiency	No Vitamin D Deficiency
Poor	53	74
Fair	160	232
Good	139	265
Very Good-Excellent	94	216

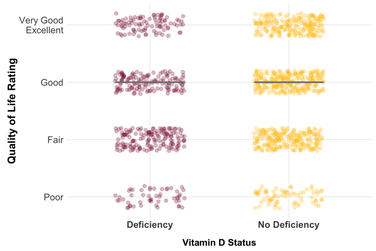

Figure 1. **Quality of Life Ranks by vitamin D Status** - Visualizes the Mann-Whitney U Test. The distribution of QoL rankings revealed a significant shift overall ($p = 0.01474$). The non-deficient group had a statistically significant tendency to rate their QoL higher.

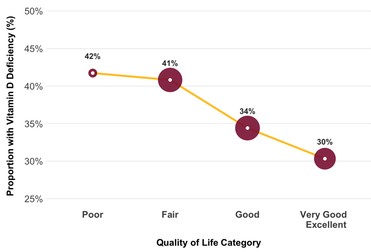

Figure 2. **Trend of Vitamin D Deficiency Across Quality of Life** - Visualizes the Cochran-Armitage Test. There was a statistically significant trend ($p = 0.0017$) showing that the presence of a VDD correlated with a lower likelihood of ranking quality of life as "Good-Excellent".

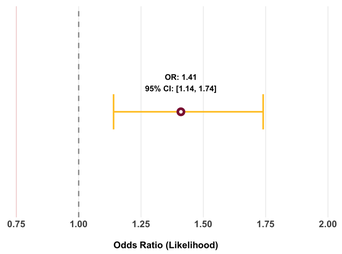

Figure 3. **Vitamin D Status and Quality of Life Odds** - Visualizes the Ordinal Logistic Regression Model. Adequate vitamin D levels are associated with a 41% increase in odds of having a higher quality of life ($OR = 1.41$, 95% CI $[1.14, 1.74]$). The t-value of 3.192 provides evidence to reject the null.

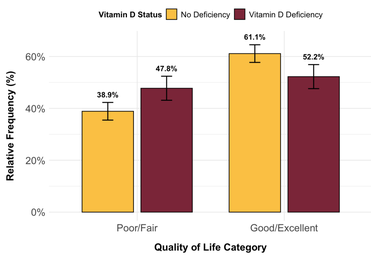

Figure 4. **Relative Frequency of Quality of Life** - Percentages calculated within each Vitamin D cohort (Each cohort = 100%). EDS patients without VDD rated their QoL more frequently as Good/Excellent and less frequently as Poor/Fair in comparison to EDS patients with VDD.

Discussion

Our results suggest that for EDS patients, the effects of Vitamin D deficiency could compound pre-existing manifestations. Symptoms such as bone fragility, chronic pain, and fatigue might intensify, further exacerbating the condition and diminishing overall quality of life. This pattern **supports existing research** showing that high-dose vitamin D supplementation correlated with decreases in VAS scores of patients with chronic musculoskeletal pain (Gendelman et al., 2015).

While we cannot claim causation here, the statistical signal is strong enough, and the clinical implications meaningful enough to warrant routine Vitamin D screening and future longitudinal studies to evaluate causation.

Impact

The 13 subtypes of EDS all have innumerable consequences on the musculoskeletal system, immune function, and energy levels, which can cause pain and greatly impact patients' QoL. Vitamin D is a vital nutrient that helps maintain the balance of calcium in the blood and bones, aiding in bone integrity and growth, muscle function and repair, nerve signaling, the immune system, and vigor.

Given that Vitamin D deficiency is easily detectable and **treatment is safe and low-cost**, recognizing whether correcting VDD would improve the QoL of this underserved population may offer a simple, accessible way to enhance affected individuals' well-being.

Acknowledgements

We would like to thank Arizona State University and the SOLUR program for further welcoming online students and supporting them in taking part in scientific research. We would also like to thank Janette Wilson for her ongoing support throughout this project. We further acknowledge the NIH and *All of Us* research program for providing the foundation of our research.

The *All of Us* Research Program is supported by the National Institutes of Health, Office of the Director: Regional Medical Centers: 1 OT2 OD026549; 1 OT2 OD026554; 1 OT2 OD026557; 1 OT2 OD026556; 1 OT2 OD026550; 1 OT2 OD 026552; 1 OT2 OD026553; 1 OT2 OD026548; 1 OT2 OD026551; 1 OT2 OD026555; IAA #: AOD 16037; Federally Qualified Health Centers: HHSN 263201600085U; Data and Research Center: 5 U2C OD023196; Biobank: 1 U24 OD023121; The Participant Center: U24 OD023176; Participant Technology Systems Center: 1 U24 OD023163; Communications and Engagement: 3 OT2 OD023205; 3 OT2 OD023206; and Community Partners: 1 OT2 OD025277; 3 OT2 OD025315; 1 OT2 OD025337; 1 OT2 OD025276. In addition, the *All of Us* Research Program would not be possible without the partnership of its participants.

SOLUR
sols.asu.edu/solur

All of Us
RESEARCH PROGRAM

Abstract

References

Github: https://github.com/xeriksen/VitaminD_EDS